THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rod is heated at one end. The time taken for wax pins to fall off is least for the rod made of: 1

(a) copper **(b)** glass **(c)** wood **(d)** iron

2. During sea breeze, the pressure of air above land is: 1

(a) the same **(b)** zero

(c) higher than over sea **(d)** lower than over sea

3. Which of these clothes would be best on a very hot, sunny day? 1

(a) leather **(b)** black polyester

(c) white loose cotton **(d)** woollen

4. Heat reaches us from a bonfire at a distance mainly by: 1

(a) conduction **(b)** radiation **(c)** evaporation **(d)** convection

5. During land breeze, warm air rises over the: 1

(a) desert **(b)** sea **(c)** land **(d)** mountain

6. Which of these would be best on a cold day? 1

(a) net cloth **(b)** layers of woollen clothes

(c) one thin shirt **(d)** sleeveless

7. When you place your hand above a candle flame, you feel heat mainly due to: 1

(a) radiation only **(b)** convection **(c)** evaporation **(d)** conduction

8. The sea is cooler than land during the day because: 1

(a) water is hot **(b)** land is wet

(c) sun does not shine on sea **(d)** water takes longer to heat

9. Two layers of thin clothes keep you warmer than one thick layer because: 1

(a) thin clothes are heavy **(b)** air gets trapped between layers

(c) no reason **(d)** thick clothes are cold

10. When you place your hand beside a candle flame, you feel heat mainly due to: 1

(a) radiation **(b)** convection **(c)** conduction **(d)** evaporation

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a metal rod conduct heat faster than a wooden stick? 2

12. Why are cooking pots with copper bottoms common? 2

13. Why do doctors use ice packs for fever? 2

14. How does a hot-air balloon rise? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how an electric heater warms a room using all modes of heat transfer. 3

16. Give reasons: 3
(a) pans are made of metal and have plastic handles
(b) the tank for hot water is covered with a wool jacket.

17. Describe the role of radiation in cooking on a charcoal grill. 3

OR

Explain why sea breeze is stronger in the afternoon. 3

18. A student heats a metal rod at the centre. Describe how heat spreads. 3

OR

Why does the handle of a mug become hot when hot tea is poured? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the three experiments to demonstrate conduction, convection and radiation. 5

OR

Describe the construction of a solar water heater and explain the heat transfer in it. 5

20. Explain how the combined effects of conduction, convection and radiation work in a thermos flask and what could happen if the vacuum is lost. 5

OR

Explain why metal vessels get hot quickly but cool down fast, compared with earthen pots. 5

END OF QUESTION PAPER — Total: 40 Marks

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